Foundations of Data Engineering and Analytics

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Sheet 4: Multivariate EDA and Dataset Comparison

In Sheet 3, you reconstructed a snapshot of DevHire Europe from its rendered website and explored single variables. Now determine how accurately you collected it, investigate relationships between variables, and compare the 2025 and 2026 snapshots.
Download the synthetic tech jobs dataset `synthetic_tech_jobs_2025_2026.csv` and its datasheet `DATASHEET.md` from OLAT. The CSV contains 2,400 fictional advertisements: 1,200 for each year. The 2026 rows describe exactly the advertisements exposed by the Sheet 3 website; we refer to this file as the *canonical data*. Read the datasheet, which describes all columns, before starting. A new crawl is not required.

About the data: The advertisements are synthetic. They were generated from the Stack Overflow Developer Surveys 2025 and 2026, and all companies and advertisements are fictional. The salary ranges are derived from what survey respondents reported as their total annual compensation; they are not base salaries advertised by employers. Required experience, seniority, technologies, and company descriptions were partly created by the generator. Seniority, for example, was derived from experience, so their relationship is built in. Each survey year has different respondents, so the 2025 and 2026 rows are not the same people or vacancies observed twice. Your findings therefore describe this generated platform, and a difference between the years does not, by itself, show a real change in the labour market.

Before you start, prepare the data: Treat empty cells as missing values, not as zeros. Derive the *salary midpoint* as in Sheet 3 and `technology_count`, the number of distinct technologies per advertisement. Apply the same steps to both years, state how you handled missing values, and report the number of observations used in each analysis.

Exercise 1 (Validation and Multivariate Analysis) [6 Points]

Use the 2026 rows of the canonical data for the multivariate analyses.

a) *1 Point* **Validate the crawl.** Join your Sheet 3 collection to the canonical 2026 rows on `job_id`. Compare the original fields; the derived variables play no role here. Compare coverage, duplicates, values, types, and missingness. Distinguish genuinely missing fields from parsing errors or missing advertisements. Explain discrepancies, including differences caused only by formatting or technology-tag ordering. Continue with the canonical data so collection mistakes do not propagate. If your crawl was incomplete or failed, discuss its limitations rather than filling it silently from the canonical data.

b) *2 Points* **Covariance.** Compute and interpret the covariance between `required_experience_years` and salary midpoint. Then display the covariance matrix that additionally includes `technology_count`. Can the covariances alone tell you which of the three pairwise relationships is strongest? Explain why or why not. Support your interpretation with appropriate plots, use a consistent set of complete observations for the matrix, and discuss units and scale.

c) [1.5 Points] **Correlation.** Compute Pearson's correlation between required experience and salary midpoint and visualize their relationship in a scatterplot. Investigate its sensitivity to removing outliers (use the outlier rule you applied in Sheet 3, e.g., Tukey's $1.5 \times \mathrm{IQR}$ rule) and to using log salary, i.e., the natural logarithm of the salary midpoint. Report which observations were removed for a sensitivity analysis; retain the unfiltered result as a reference. Explain why association does not establish causation and why correlation alone does not describe the whole relationship.

d) [1.5 Points] **Relationships across categories.** Compare the salary–experience relationship across role families using grouped scatterplots and correlations. Display salary distributions across seniority levels and compare their outlier counts *and proportions* using the same stated rule. Also report mean required experience by seniority. Interpret group sample sizes, possible confounding, and the built-in link between seniority and experience.

Exercise 2 (2025 versus 2026: Dataset Comparison) [4 Points]

Use the data prepared above for both years, with the same plotting scales and choices for each year. Both snapshots use the same generator version and sample size, but their source respondent populations and composition can differ.

a) [1.5 Points] **Shape.** Compare the salary-midpoint and required-experience distributions using suitable overlaid or aligned plots and numerical summaries. Discuss location, skewness, tails, and outliers. Are the differences still visible when you compare the same role family in both years?

b) [1.5 Points] **Spread.** Compare variance, standard deviation, and interquartile range for salary midpoint and required experience. Explain differences between robust and non-robust measures. Examine whether differences in country or role composition could contribute to the observed changes; state group sizes.

c) [1 Point] **Relationships and defensible conclusions.** Compare the covariance and correlation between required experience and salary midpoint in both years. Summarize one supported finding and one tempting but unsupported claim. Distinguish source-survey differences, sample composition, generator transformations, and genuine market change. Explain what additional data would be needed to assess the latter.

Submission. Submit the reproducible `.ipynb`, any separate analysis source, dependency instructions, and an exported `.html` or `.pdf` version to OLAT. Include crawl-validation results, calculations, charts, sample sizes, and interpretations. Also include the Sheet 3 collection you validated, so that the validation can be reproduced. There is no need to resubmit the provided CSV or rerun the crawler.

Important: Submit your solution to OLAT and mark your solved exercises with the provided checkboxes. The deadline ends at 23:59 on the day before the discussion.